

Machine Learning Basics: Supervised vs Unsupervised Learning

What Is Machine Learning?

Machine Learning (ML) is a field of artificial intelligence where computers learn patterns from data and make decisions or predictions without being explicitly programmed.

- Recognize patterns
- Make predictions
- Improve over time

Supervised Learning

Supervised Learning uses labeled data, where each input has a known output.

- Uses labeled data
- Focuses on predictions
- Tasks: classification, regression

Examples of Supervised Learning

- Spam Detection: classify emails as spam or not
- House Price Prediction: predict price from features

Unsupervised Learning

Unsupervised Learning uses unlabeled data to find hidden patterns.

- Uses unlabeled data
- Finds patterns
- Tasks: clustering, association

Examples of Unsupervised Learning

- Customer Segmentation: group users by behavior
- Recommendation Systems: suggest similar content

Key Differences

Feature	Supervised	Unsupervised
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Data	Labeled	Unlabeled
Goal	Predict	Find patterns
Output	Known	Unknown

Conclusion

Supervised learning predicts outcomes using labeled data, while unsupervised learning discovers hidden patterns in unlabeled data.